

Deep Insights into the Drift Diffusion Model: From Basics to Variants

Yong-Ming Tian, Zong-Yi Chen, Shiqi Liu, Yajie Li

Summary:

Past decades have witnessed a great interest in Drift Diffusion Model (DDM) in the fields of neuroscience [1][2][3], which succeeded in modeling numerous decision-making processes and has plenty of variants. Recently, during the craze of Xiao Yuan Kousuan, we made decisions on comparing numbers and compared our accuracy and response speed. This sparked our interest in humans' decision-making process, and we wanted to explore DDM to better understand it. Specifically, we explored **the basic DDM** and also **its variants**, all using a simple but robust size-comparison task, and **all inspired by real-life thoughts**. In our project, we delved more into DDM, gained better insights of how individuals make decisions under varying conditions, such as with time pressure or pre-existing biases. Most of the experiments have good results and can be modeled and explained by basic DDM or variants of DDM pretty well. The others (mainly in Section 4), although yielding results that were **initially unexpected**, have been thoroughly analyzed after our deep thinking.

1 Preliminaries

1.1 The Drift Diffusion Model

The Drift Diffusion Model (DDM), commonly used in the fields of psychology and neuroscience, describes two-choice decision-making as a process of accumulating evidence over time until it reaches one of the two preset thresholds, at which point a choice is made. The model includes parameters such as drift rate (evidence accumulation speed), boundary separation (decision threshold), starting point (initial bias), non-decision time (delays unrelated to accumulation), and noise (random fluctuations). The measures of performance are typically **accuracy**, **response times (RTs)** and **the probabilities of making the two choices**, typically vary with coherence.

1.2 Basic Experimental Setup

We designed all our experiments as shown in Figure 1, which described a simple size-comparison task: In each trial, the subject should fix their gaze on the center cross "X", then use peripheral vision to judge which of the two solid circles on the left or right is larger, and press the corresponding key to indicate the larger circle. In different trials, the basic size of the left circle is randomized, and the variable in this experiment is the size ratios between the two circles designed as $r = (\text{size_right}/\text{size_left}) - 1$. The mean of r is 0, with values greater than 0 indicating that the right circle is larger, and the magnitude of the value reflecting the degree of difference. And the values less than 0 can be explained conversely. In this context, r can be understood as "coherence" in "random dots motion experiments" firstly proposed by Britten *et al.* [4], and set to be values in $[-0.32, -0.24, -0.16, -0.08, 0, 0.08, 0.16, 0.24, 0.32]$ in our experiments. The subjects are instructed to **prioritize accuracy** while also making their decisions as **quickly** as possible.



Figure 1: Experiment demo

2 The Basic DDM Experiment

15 undergraduates of **Nanjing University** and **Tsinghua University** were invited to do the basic DDM experiment described in Section 1.2 (10 for variants of DDM). Each of them completed 160 size-comparison trials to ensure a substantial dataset, with trials evenly distributed across the different ratios. We analyzed these experimental results, then plotted the accuracy, probability of choosing the right option, and response times as functions of the ratios r , finding that the results closely followed the patterns of the DDM. Finally we employed a method called grid-search, commonly used in machine learning, to find the suitable parameters of DDM to best fit our basic experimental results. The goal of the fitting was to minimize the mean squared error between the RT- r curve generated by the DDM and that of the aggregated real experimental data from all participants. Figure 2 provides a visualization of the results from different subjects. Figure 3 shows the effectiveness of our grid-search method. Figure 4 displays that our results can be well-fitted by DDM with the best parameters. The figures showed clear results and was easy to understand (skipped here for space). However, we remembered that there was a **red** in RT- r curve of DDM in our homework, while in our basic DDM Experiment, not. **Why?** The subjects consumed too much time to judge when r is 0, and we need **time pressure!** This is how we come to design Variant I.

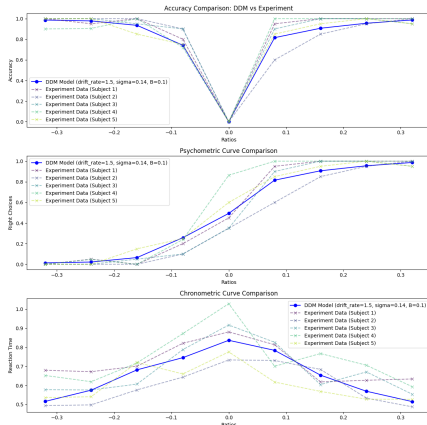


Figure 2: Different subjects' results

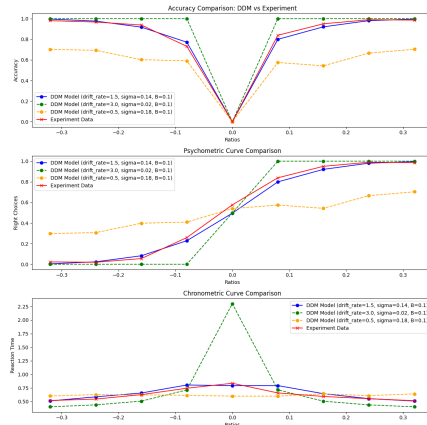


Figure 3: Fitting by different parameters

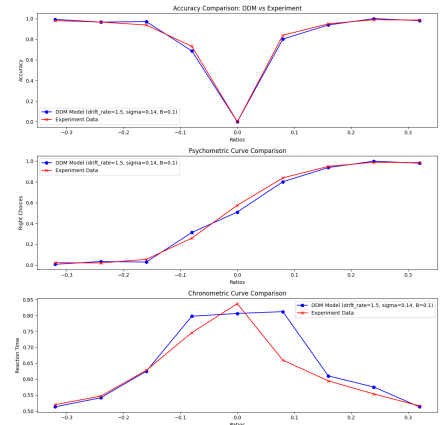


Figure 4: The best fit

3 The Variants of DDM Experiment

3.1 Under High Time Pressure–Collapsing Bound

In Variant I, we introduced high time pressure, by emphasizing the need for speed in advance, and also displaying a prominent “Faster” prompt on the screen if a subject’s response is too slow during any trial. With this change, accuracy slightly decreased, response times decreased overall and became closer to each other across different ratios. Interestingly, the plot of response times versus ratios showed a **dent** in the middle, meaning when the size of the circle was the same, the subjects were more likely to **guess** an answer more quickly. When fitting the experimental results, we found that the suitable boundary parameter is significantly smaller, which is consistent with the paper[5]. Sadly, the basic DDM didn’t fit the experiment well, but we found a more satisfying solution—using the Collapsing Bound DDM. We employed a boundary that decreases exponentially over time, and the results were pretty well, with a mean square residual in RT- r curve of **0.000262**, **one third** of that of the basic DDM. Even more surprisingly, the fitting curve in RT- r curve also showed a **dent** in the middle. **Why?** To explain it, we plotted the accumulation of evidence as Figure 5, and the stone is out of water! The collapsing bound makes it quite easier for evidence accumulation to reach the boundary when r is 0, due to the pure influence of noise. On the other hand, in cases where r is not zero but has a quite small absolute value, it may take longer to reach the boundary.

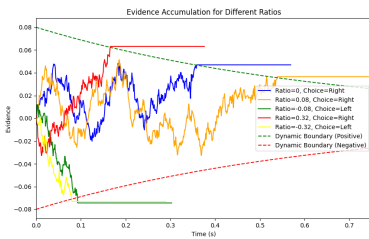


Figure 5: (Variant I) Accumulation of evidence-collapsing bound

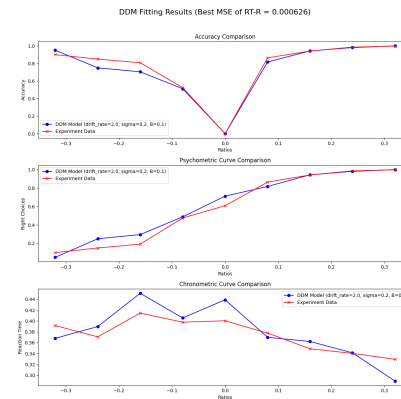


Figure 6: (Variant II) The results and fitting

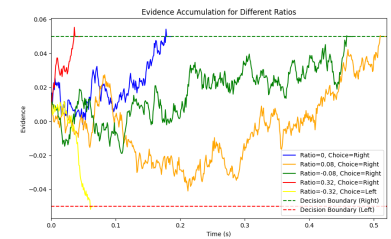


Figure 7: (Variant II) Accumulation of evidence-initial bias

3.2 With Asymmetric Prior–Change in the Starting Point of Evidence Accumulation

We noted the **symmetry** assumption in the basic DDM, which meant no prior with bias could be considered. However, when we judge something or somebody, we always have our own priors, often asymmetrical. Also **inspired by our teacher**, we designed Variant II, where we explored the influence of the asymmetrical priors on DDM. We split the experiment into two parts. In one part, the circle on the right was bigger with a probability of 75%, while in the other, reversed. At the beginning of each experiment, the probability **was** explained to the subjects. Since the results of the two parts turned out to be completely opposite, we focused our analysis on the condition where the probability of the right circle being larger was higher. The results were amazing as shown in Figure 6. A lot of asymmetry was observed: When r was bigger than 0, namely the right circle was bigger, the accuracy was higher while the reaction time was lower. Also, the subjects’ probability of choosing right significantly increased. The basic DDM obviously could not fit the experiment well, so we changed the starting point of the evidence accumulation, about 1/3 closer to the boundary of the right, inspired by the paper [5]. The probability of reaching the boundary of the right thus increased as Figure 7 showed, and the blue curve in Figure 6 showed this variant of DDM displayed similar qualities and fit the experimental results well. As the results shows, **the prior influences us really a lot!** **A vivid example in our life** is people tend to see the character on the left side of a poster as good while the right as bad.

3.3 Apply Visual Disturbances– Adding Considerable Noise

While doing previous experiments, we once obtained **a terrible data** made by my roommate. And the reason was during his trials, I was **talking to him** all the time! So we got the idea to study the influence of “noise” on our decision processes, which gives birth to Variant III. Here, we employed an interesting way to add random noise to the experiment. Specifically, we added a circular background frame with randomized size around the circle as Figure 8 shows. Compared to the experiment with-



Figure 8: Noised experiment demo out noise (**all** the difference is with or without noise), the noised experiment resulted in lower accuracy and longer reaction times. As for the optimal parameters that fit the experiment, the noise parameter σ of the DDM increased significantly, from **0.22** to **0.28**, while the drift rate decreased significantly, from **2.25** to **1.3**, which could be explained by the decrease of confidence influenced by the added noise.

4 More Extensions of DDM and A deeper insight

We also explored other extensions of DDM, such as DDM with Working Memory Task, DDM with Sequences of ascending or descending ratios. We have gained a deeper understanding of this model and decision-making behavior in neuroscience, and our capabilities have significantly improved. Due to space limitations, these parts of our project will only be shown **in the presentation** later.

5 Conclusion

Our project are **not task-oriented but thinking-oriented**. We studied DDM and many variants of DDM, all inspired by interesting and natural thoughts. In each experiment, we performed qualitative and quantitative analyses, and did our utmost to explain and fit the experimental results well. We have **a lot more to display** and **eagerly want to do the presentation**(we’ve already made the power-point).

Division of Labor

All the members in our team put in **one hundred percent** effort. The **main** division of labor is listed as follows, but only serve as a reference, as we **all** participate in **all** the processes of the **whole** project.

- **All the team members**, actively participated in discussion of the project idea, all wrote some parts of the core codes and did some parts of the plotting to ensure that **everyone is deeply involved** in the project, all responsible for collecting experimental data.
- **Yong-Ming Tian 221900180**, tianym2022@smail.nju.edu.cn, the leader, gave the idea, coordinated the division of labor, provided basic guidance on various aspects of the entire project, designed, tested and fit the basic experiment, did the writing.
- **Zong-Yi Chen 221900171**, chenzy2022@smail.nju.edu.cn, designed and tested Variant I DDM experiment and Variant II DDM experiment, explored the more extensions of DDM, broadened our horizon of DDM and the decision processes in neuroscience.
- **Yajie Li 221900331**, 221900331@smail.nju.edu.cn, designed and tested Variant III DDM experiment and fit Variant I DDM experiment and Variant II DDM experiment, contributed to the refinement of our project in various aspects.
- **Shiqi Liu 221900136**, 221900136@smail.nju.edu.cn, fit Variant III DDM experiment, read plenty of academic paper, provided deeper insights and analysis, summarized everyone's gains of the project, made the PowerPoint and will do the presentation.

References

- [1] D. Fudenberg, W. Newey, P. Strack, and T. Strzalecki, "Testing the drift-diffusion model," *Proceedings of the National Academy of Sciences*, vol. 117, pp. 33141–33148, Dec. 2020.
- [2] C. E. Myers, A. Interian, and A. A. Moustafa, "A practical introduction to using the drift diffusion model of decision-making in cognitive psychology, neuroscience, and health sciences," *Frontiers in Psychology*, vol. 13, p. 1039172, Dec. 2022.
- [3] L. G. Alvarez-Zuzek, J. Grujic, and R. Gallotti, "Drift Diffusion Model to understand (mis)information sharing dynamic in complex networks," Jan. 2024. arXiv:2401.17846 [physics].
- [4] K. Britten, M. Shadlen, W. Newsome, and J. Movshon, "The analysis of visual motion: a comparison of neuronal and psychophysical performance," *Journal of Neuroscience*, vol. 12, no. 12, pp. 4745–4765, 1992.
- [5] R. Ratcliff and G. McKoon, "The Diffusion Decision Model: Theory and Data for Two-Choice Decision Tasks," *Neural Computation*, vol. 20, pp. 873–922, Apr. 2008.

Supplement

This part shows our additional work, without limitation of page.

S.1 Extension I Experiment

In Extension I, we examined the impact of memory load on decision-making tasks. We asked the subjects to remember numbers, which **continuously appeared** in sequence during the experimental process. The reason why we didn't ask them to just remember a series of numbers at the beginning of the experiment was that it would impose less memory load, as the subjects would have memorized the constant numbers deeply as the experiment progressed. After each serials of trial, we would ask the subjects about the specific position of certain numbers in the sequence, and low accuracy rate would render this serial of trial invalid. As we expected, the accuracy rate of the subjects decreased. However, the reaction time changes unexpectedly. We originally hypothesized that reaction times would increase as subjects took additional time to remember the numbers, but the results turned out to be the opposite. We analyzed this and got the answer, and here is an interesting analogy: When we take political exams, our writing speed is often faster than usual because, over time, we might forget the content we have memorized. So the memory load can be seen as another way of time pressure!

S.2 Extension II Experiment

In Extension II, we let the absolute value of the ratios $|r|$ appear in an ascending or descending order in each series of trial. The results of reaction time differed greatly, as shown by Figure 9. When $|r|$ appeared in an descending order, the RT- r curve was obviously "broader". **Why?** In this case, the task became more and more hard, and the subjects' confidence became more and more lower. They consequently needed more concentration to tackle the tasks. So the curve changed greatly with $|r|$. In the other case, the subjects found the experiment was hard from the beginning, and they displayed enough concentration. When the difficulty decreased, the concentration would not lose that quickly, causing the curve to be more "smooth". This are our explanation, might wrong, but with rationality and deep thinking.

S.3 Deeper Insights

We have gained deeper insights of the project, these will be shown in the presentation. Here is a brief one:

1. deeper understandings of the parameters of DDM:

We gained deeper insights of how the parameters influence a DDM's behavior. A interesting question here is when we did the grid search for the best parameters, we searched for three—drift rate, bound and noise. However, in the expression of DDM, they seemed to have a nearly linear relation. A bigger drift rate plus a lower noise naturally equal to a lower bound. Unluckily, when we only search for two parameters, the effect was not that ideal, perhaps due to the randomness in the DDM simulation, so their relation was not fully linear.

2. Shortcomings of DDM:

Despite the success of DDM, we have found some articles on the Internet criticizing the model, such as its relatively strict conditions to use in real life. This part greatly broadened our horizon.

3. Other Models about decision:

There were more models about decision, such as Decision Tree and Bayesian Method. We compared DDM and them, and gained better understandings.

4. Inspirations for Our Life:

East or West, inspirations for life is best. Our project idea came from life, and we finally gained inspirations for our life, such as how to understand merits and demerits of human decision, how to make good decisions and view others' decisions and etc.

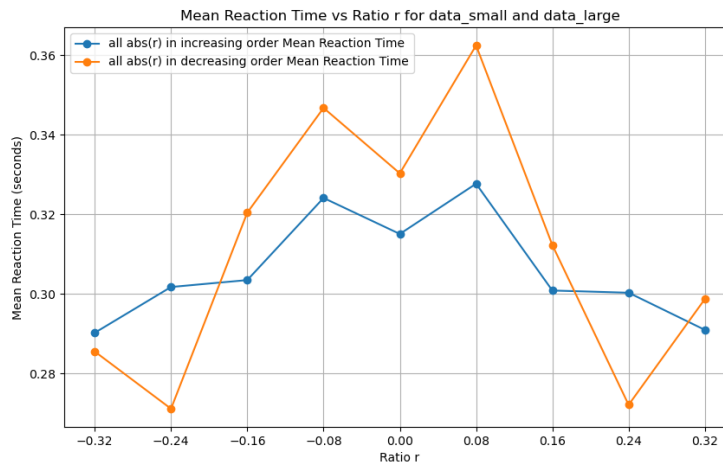


Figure 9: (Extension II) The results

S.4 Plot With Error Bar

We thank our teacher for pointing out the laxity of our statistical plot. So in this section, we provide a plot with error bar on the reaction times with various r in the basic DDM Experiment, as Figure 10. The effect is ideal. The error bar plot shows the variance of 1000 trials of our simulation of DDM. When $r = 0$, the model has the biggest variance, similar to humans when judging the size of two circles of the same size. And when $|r|$ is quite large, the variance is pretty small, as the speed of the accumulation of evidence is so fast, mitigating the effects of randomness, noise, and other factors. The figures of other experiments can be plot with error bar similarly.

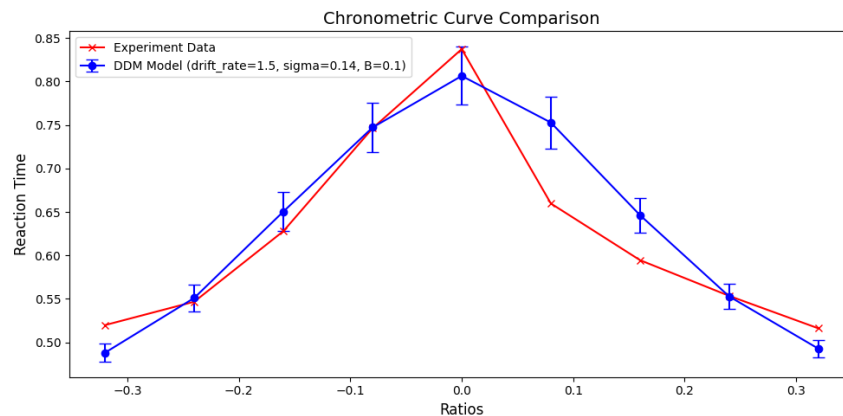


Figure 10: Plot with error bar